

QUANTUM MECHANICS FROM THE JOBSON CELL MEDIUM

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FRAMING NOTE (added 2026-08-27, v4):

The photon descriptions in this document are quantitatively correct. The correct qualitative picture, developed in subsequent work, is that a photon is a localized T_{1g} wave packet (corpuscle, Newton's term) -- not an extended sinusoidal wave. What propagates as "light" is the T_{1g} pressure wave in the medium; the photon corpuscle is the quantized, localized form of that excitation. Wave-particle duality in standard QM arises from conflating the medium wave with the localized packet. Note: T_{1g} waves may also be generated by macroscopic charge oscillations (classical EM) without individual corpuscle events -- whether all T_{1g} propagation requires a corpuscle source is not yet derived. [OPEN] An electron is a corpuscle locked into a stable (1,2) Hopf winding; the winding gives structure to its medium wave. A photon corpuscle carries no persistent winding ($\theta=0$). Both are corpuscles; the winding is the only difference. All mathematical results in this document are unchanged. The quantized, corpuscular nature of light -- established by the photoelectric effect -- is consistent with localized T_{1g} wave packets. Newton lacked the medium to explain the accompanying wave behavior.

ABSTRACT

The Schrödinger equation is not a postulate. It is the non-relativistic limit of the Klein-Gordon equation that governs Hopf winding modes in the Jobson cell medium. The derivation chain has zero free parameters:

1. Jobson medium: $K = 1/\epsilon_0$, $\rho = \mu_0$, $c = \sqrt{K/\rho}$ [PROVEN]
2. Free pressure wave: $\omega = c*k$ (massless) [QM1]
3. Hopf winding rest energy = $m_p*c^2 \rightarrow$ Compton frequency $\omega_C = m_p*c^2/\hbar$ [QM2; the Compton wavelength IS $\lambda_p = \hbar c/m_p$ = Zone 1 boundary]
4. Winding dispersion: $\omega^2 = c^2*k^2 + \omega_C^2$ [Klein-Gordon, QM3]
5. Non-relativistic limit ($v \ll c$):
 $\omega_{KG} \rightarrow \omega_C + \hbar*k^2/(2*m_p)$
Remove rest-mass oscillation $\phi = \psi * \exp(-i*m_p*c^2*t/\hbar)$:
 $i*\hbar * d\psi/dt = -(\hbar^2/2*m_p) * \nabla^2 * \psi$ [SCHRÖDINGER, QM5]
6. Born rule: $P(x) = |\psi(x)|^2$ from medium wave energy density [QM6]

The Schrödinger equation is quantum mechanics. It is now derived from the same medium (Jobson cell lattice) that gives α , G , the Higgs mass, and quantum entanglement. No additional postulates or free parameters.

NEW GROUND: no existing interpretation of quantum mechanics -- Copenhagen, many-worlds, de Broglie-Bohm, GRW, QBism, consistent histories -- derives the Schrödinger equation, the Born rule, or the Tsirelson bound from an underlying physical medium with independently fixed mechanical properties. Every existing approach postulates the Hilbert-space/probability structure as an axiom. This paper derives it instead, as a mechanical consequence of the same medium that gives α , G , and the Higgs mass in the companion papers.

Consequences:

- Double-slit: Hopf winding fields span both slits simultaneously.
Interference = constructive/destructive Huygens superposition in medium.

- Which-path: Zone 3 field decoherence resolves winding at $r < r_{\text{lock}}(T)$.
 $r_{\text{lock}}(300\text{K}) = 340 \text{ fm}$; coherence preserved for macroscopic slit separations.
- Delayed choice: no paradox. Medium configuration never had a single-slit address. "Measurement" = local perturbation resolving global winding.
Timing is irrelevant; topology is not retroactively changed.

NEW PREDICTIONS (not in standard QM):

- Minimum electron slit width for coherent diffraction: $2\lambda_{\text{bar}_e} = 772 \text{ fm}$
(below this, slit-edge Zone 3 fields decohere the electron winding)
- Minimum proton slit width: $2\lambda_{\text{bar}_p} = 0.421 \text{ fm} = r_{\text{grind}}$ (nuclear hard core)
- Fringe visibility vs temperature: $V(T) \sim \exp(-r_{\text{sep}} / r_{\text{lock}}(T))$
where $r_{\text{lock}}(T) = (\alpha \hbar c^2 / k_B T)^{1/3}$

Companion script: analysis/demos/qm_doc.py (21/21 PASS)

1. MEDIUM WAVE EQUATION AND KLEIN-GORDON

1.1 Free pressure waves

The Jobson cell medium has bulk modulus $K = 1/\epsilon_0$ and density $\rho = \mu_0$.

The pressure wave speed $c = \sqrt{K/\rho} = 1/\sqrt{\epsilon_0 \mu_0}$ (exact by SI).

Free waves satisfy:

$$(d^2/dt^2 - c^2 \nabla^2) \phi = 0$$

Dispersion: $\omega = c \cdot k$ (massless, linear). [QM1]

1.2 Hopf winding rest energy -> mass gap

A Hopf winding mode (the proton) is a topological defect in the Jobson cell medium with winding number (1,2) ($p=1, q=2$). The Maxwell jamming constraint ($N_J = 21$ cells, $3V-E = 6$ exactly critical) sets the minimum size of the winding. Below this size, the winding cannot be compressed -- this creates a minimum energy:

$$E_0 = m_p \cdot c^2 \quad [\text{proton rest energy} = \text{mass gap of the winding mode}]$$

This generates the Compton oscillation frequency:

$$\omega_C = m_p \cdot c^2 / \hbar$$

Crucially: the Compton wavelength $\lambda_{\text{bar}} = \hbar / (m_p \cdot c) = \hbar_c / m_p$

IS the Zone 1 boundary λ_p . The winding resonates at the Compton frequency WITHIN Zone 1 (the Maxwell critical jamming region). This is the same λ_p that appears in $r_p = 4\lambda_p$ (PS4) and throughout nuclear structure.

The identification $\lambda_{\text{bar}} = \lambda_p$ is not a coincidence: it follows from the proton being the Maxwell-critical Hopf winding in the Jobson cell medium.

[QM2 PASS: $\omega_C = c / \lambda_p$ confirmed to machine precision]

1.3 Klein-Gordon equation for the winding mode

With a mass gap, the dispersion relation becomes:

$$\omega^2 = c^2 \cdot k^2 + \omega_C^2 \quad [\text{Klein-Gordon}]$$

At $k=0$ (standing winding): $\omega = \omega_C$ (pure Compton oscillation)

At $k \gg \omega_C/c$: $\omega \rightarrow c*k$ (ultra-relativistic, massless limit)

This IS the Klein-Gordon equation: $(\partial^2/\partial t^2 - c^2 \nabla^2)\psi = -(m_p c^2/\hbar)^2 \psi$

[QM3 PASS: both limits verified]

2. NON-RELATIVISTIC LIMIT: SCHRÖDINGER EQUATION

For non-relativistic motion ($v \ll c$, $k \ll \omega_C/c = m_p c/\hbar$):

$$\begin{aligned}\omega_{KG} &= \sqrt{c^2 k^2 + \omega_C^2} \\ &= \omega_C * \sqrt{1 + (\hbar k / m_p c)^2} \\ &\sim \omega_C + \hbar k^2 / (2 m_p) + O((v/c)^4)\end{aligned}$$

The ω_C term = rest-mass oscillation. Extract it:

$$\psi(x,t) = \phi(x,t) * \exp(-i * m_p c^2 t / \hbar)$$

Substituting into Klein-Gordon and dropping the $O((v/c)^2)$ correction to $\partial^2 \psi / \partial t^2$:

$$i \hbar \frac{d\phi}{dt} = -(\hbar^2 / 2 m_p) * \nabla^2 \phi$$

This IS the time-dependent Schrödinger equation for a free particle. [QM5 PASS]

Accuracy: The approximation is exact to order $(v/c)^2$. For $v/c = 1/1000$ (deep NR):

relative error = $2.5e-7$, consistent with the $(v/c)^2/4$ Taylor expansion.

ZERO additional assumptions were made. The Schrödinger equation follows from:

- (a) The Jobson cell medium (K , ρ , c -- already proven)
- (b) The Hopf winding mass gap $E_0 = m_p c^2$ (from Maxwell jamming -- already proven)
- (c) The non-relativistic approximation (standard physics -- no extra postulate)

Potentials ($V \neq 0$): adding an external pressure field $V(x)$ to the medium gives:

$$i \hbar \frac{d\psi}{dt} = -(\hbar^2/2m) \nabla^2 \psi + V(x) \psi$$

This is the full Schrödinger equation. $V(x)$ = the local Jobson cell pressure

(Coulomb, gravity, nuclear) -- all derived from the same medium.

3. BORN RULE FROM WAVE ENERGY DENSITY

In the Jobson medium, the energy density of a pressure wave is:

$$u(x,t) = (1/2) \rho v^2 + (1/2) K \text{strain}^2 \propto |\psi(x,t)|^2$$

The probability of a detector at x absorbing the wave (detecting the particle)

is proportional to the wave energy density at x :

$$P(x) = u(x) / \int u \, dV = |\psi(x)|^2 / \int |\psi|^2 \, dV \quad [\text{Born rule}]$$

This is the Born rule. It requires no additional postulate: detection = energy

absorption from the medium wave, and energy density is proportional to amplitude squared.

[QM6 PASS: Born rule from medium energy density]

Note on wave identity: Light propagation = T_{1g} pressure wave in the Jobson

cell medium (Klein-Gordon with $m=0 \rightarrow$ Maxwell equations). A photon is a

localized T_{1g} wave packet (corpuscle) that IS one source of such waves;

macroscopic charge oscillations (classical EM) may generate T_{1g} waves through

the cog/Zone 3 mechanism without individual corpuscle emission events. Whether

all T_{1g} waves require a corpuscle source or can be generated purely classically is not yet derived from the framework. [OPEN: classical vs quantum T_{1g} generation -- Series 3 target.] Matter (electrons, protons) = massive Hopf winding modes (Klein-Gordon with mass gap m^*c^2). Light waves and matter windings are two excitation types of the same medium, not separate entities. The photon (corpuscle) is the $\theta=0$ mode. Here θ is the winding tilt angle: it measures how far the Hopf winding departs from the massless T_{1g} baseline. $\theta=0$ means no Zone 1 displacement and no rest mass; $\theta>0$ means the winding displaces Zone 1 and acquires rest energy m^*c^2 . Formula: $\theta(m) = \arcsin(8\alpha\phi m/m_p)$. All matter requires $\theta > 0$.

Pair production: when a photon's pressure wave amplitude exceeds the threshold for nucleating a new Hopf winding ($E \geq 2m_e c^2$ for e^+e^- pair), the medium forces a new topological winding into existence. This is the Schwinger mechanism: field amplitude > winding nucleation threshold.

Environments where this occurs: gamma-ray burst cores ($kT \sim \text{MeV}$), neutron star merger ejecta, the early universe ($T > 5e9 \text{ K} \sim 0.4 \text{ MeV}$), and extreme laser fields approaching the Schwinger critical field ($E_S = m_e^2 c^3 / e \hbar$). Note: the solar core ($kT \sim 1.3 \text{ keV}$) is $\sim 800\times$ below threshold and does NOT continuously pair-produce. [See doc_nucleus Zone 2: $N_J=21$ for the winding nucleation condition; open_items.txt F-12 for the photon-trapping calculation]

4. DOUBLE-SLIT EXPERIMENT

4.1 Why interference occurs

The proton (or electron) Hopf winding mode in the medium has a Zone 3 field extending to infinity (falling as $1/r^2$). When the winding approaches a barrier with two slits:

- (a) The winding CANNOT be "at slit A" or "at slit B" -- it is a topological configuration of the medium, defined globally (same as the A_g singlet in doc_entanglement).
- (b) The medium pressure field of the winding passes through BOTH slit openings simultaneously.
- (c) At the screen, the pressure wave contributions from both slits superpose: constructive where path difference = $n\lambda_{dB}$, destructive otherwise.
- (d) Interference pattern = standard Huygens-Fresnel diffraction for the medium pressure wave. Fringe spacing = λ_{dB} / d where d is slit separation. $\lambda_{dB} = 2\pi\hbar / (m^*v) = \text{de Broglie wavelength (from KG dispersion)}$.

4.2 Why which-path measurement destroys interference

From doc_entanglement EP3/EP6: a detector at one slit places a Zone 3-active particle (detector atoms) within the phase-locking radius $r_{\text{lock}}(T)$ of the beam. The Zone 3 field coupling resolves the global winding into a definite path.

$$r_{\text{lock}}(300\text{K}) = 340 \text{ fm} \quad [\text{Phase-lock threshold at room temperature}]$$

For macroscopic slits (separation $\gg 340$ fm), the detector itself must physically interact with the beam to resolve the path. Simply placing a detector nearby does NOT decohere (consistent with observation: detection requires interaction).

4.2a "Sneakily turning off" the detector does not restore interference

A crucial implication: decoherence is caused by PHYSICAL ZONE 3 COUPLING between the particle's winding and detector atoms, NOT by whether a human reads the output.

Experiment: a which-path detector is "turned off" (recording stopped) but the detector atoms remain physically in place near the beam path. The particle pattern (two bands, no interference) persists.

Explanation:

Detector ON and recording: atoms within r_{lock} $\rightarrow$ winding resolved $\rightarrow$ particle pattern
Detector "off" (not recording) but atoms present: atoms STILL within r_{lock}
 $\rightarrow$ Zone 3 coupling STILL occurs $\rightarrow$ winding STILL resolved
 $\rightarrow$ particle pattern persists. "Off" means nothing.
Detector physically removed: no atoms within r_{lock} $\rightarrow$ no coupling $\rightarrow$ interference returns.

The particle's winding does not "know" whether a human is reading the detector. It only responds to whether another Zone 3 field is within 340 fm. The threshold is $r_{\text{lock}}(T)$ -- a physical scale, not an epistemic one.

This resolves a longstanding confusion in quantum measurement theory: measurement = physical interaction at $r < r_{\text{lock}}(T)$, not observation. Copenhagen is vague on this distinction. Torsionverse is not.

ALSO EXPLAINS: the quantum eraser (Kim et al. 2000). Pairs whose idler photon went through the eraser (no Zone 3 coupling forced) retain the A_g singlet $\rightarrow$ their sub-ensemble shows interference in coincidence counts. The "recovery" of the pattern is not retroactive: those pairs were never decohered to begin with.

4.3 New prediction: minimum slit width

Below $2\lambda_{\text{bar}}$ (reduced Compton wavelength), the slit-edge Zone 3 fields of slit atoms become comparable to the winding's Zone 3 field $\rightarrow$ decoherence.

Electron: $2\lambda_{\text{bar}_e} = 2\hbar c/m_e = 772$ fm [QM7]
Proton: $2\lambda_{\text{bar}_p} = 2\hbar c/m_p = 0.421$ fm = r_{grind} [QM8]

Standard QM has no minimum slit width. Torsionverse predicts one, testable with sub-pm electron diffraction experiments.

5. DELAYED CHOICE AND QUANTUM ERASER

5.1 Delayed choice: paradox dissolved

Wheeler's delayed choice: after the particle passes the slits, the experimenter decides whether to measure which-path or allow interference.

In the torsionverse:

- The "particle" is a Hopf winding mode in the medium
- After passing the slits, the medium configuration spans BOTH slit openings
- There is no "particle at slit A or B" -- only a global medium configuration
- The choice of measurement = whether to apply a Zone 3-resolving perturbation

The timing of the perturbation is irrelevant:

- "Before": resolve now $\rightarrow$ no interference (winding localized)
- "After": resolve after slits $\rightarrow$ medium was never in a single-slit state

- The resolution perturbs the CURRENT medium state, not a past state
- No retrocausality: the medium is Markovian (current state determines future)

[QM9 PASS: no retrocausality required]

5.2 Quantum eraser

When which-path information is "erased" (the resolving perturbation is undone before it propagates), the A_g global winding was never permanently resolved.

The winding continues in its global configuration. Interference returns.

This is not mysterious: the perturbation must propagate classically at c to affect the screen. If it is undone first, it never reaches the screen.

[Consistent with quantum eraser experiments by Kim et al. 2000]

6. STATUS SUMMARY

PROVEN (from existing medium derivations + this document):

- Schrodinger equation from Klein-Gordon NR limit [QM4-5, 10/10 PASS]
- Born rule from medium wave energy density [QM6]
- Double-slit interference mechanism (Huygens in medium)
- Delayed choice: no retrocausality [QM9]

ESSENTIALLY DERIVED (borrowed from entanglement_doc):

- Which-path decoherence at $r < r_{\text{lock}}(T)$ [entanglement_doc EP3; essentially derived -- t in MeV open; QM10 check is $r_{\text{lock}} > 0$ sanity only]

NEW PREDICTIONS (physical argument from Zone 3 geometry; not yet experimentally tested):

- Minimum electron slit width $2\hbar c/m_e = 772 \text{ fm}$ [QM7 -- script check verifies formula; labeled 'NEW PREDICTION' in script; Zone 3 argument]
- Minimum proton slit width $2\lambda_p = r_p/2 = r_{\text{grind}}$ [QM8 -- second condition IS a real check: $r_p = 4\lambda_p$ (PS4); first condition is formula-only]

ESSENTIALLY CLOSED:

- Full Schrodinger equation with potential $V(x)$: V = local medium pressure (Coulomb, gravity all derived from same medium; adding V is direct)
- Quantum eraser: perturbation propagation bounds the erasure window
- Pauli equation: from Dirac NR limit + $E+$ spinor [qm_dirac_pauli.py QD6]
- Dirac equation: from Clifford (T_{1g} generators) + $E+/E-$ ($2I$) [QD1-QD4]
 $g=2$ from $(\sigma.v)^2 = |v|^2$ [QD5]; proton $g_p \neq 2$ from Zone structure
- Path integral: from medium Green's function (C7 static limit = Coulomb)
Composition rule, classical action phase, $H_{E_1} = -13.6 \text{ eV}$ [QP1-QP5]

GENUINELY OPEN:

- Fringe visibility $V(T)$ quantitative formula from $r_{\text{lock}}(T)$
- Relativistic QM with spin: Dirac + EM = QED (next paper)

7. COMPANION SCRIPT

analysis/demos/qm_doc.py -- 21/21 PASS [STANDALONE -- no external deps]

Working scripts:

analysis/quantum/qm_from_medium.py -- 10/10 PASS [KG -> Schrodinger, Born rule]
analysis/quantum/qm_dirac_pauli.py -- 7/7 PASS [Clifford, Dirac, Pauli, $g=2$]
analysis/quantum/qm_path_integral.py -- 5/5 PASS [propagator, composition, E_1]

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